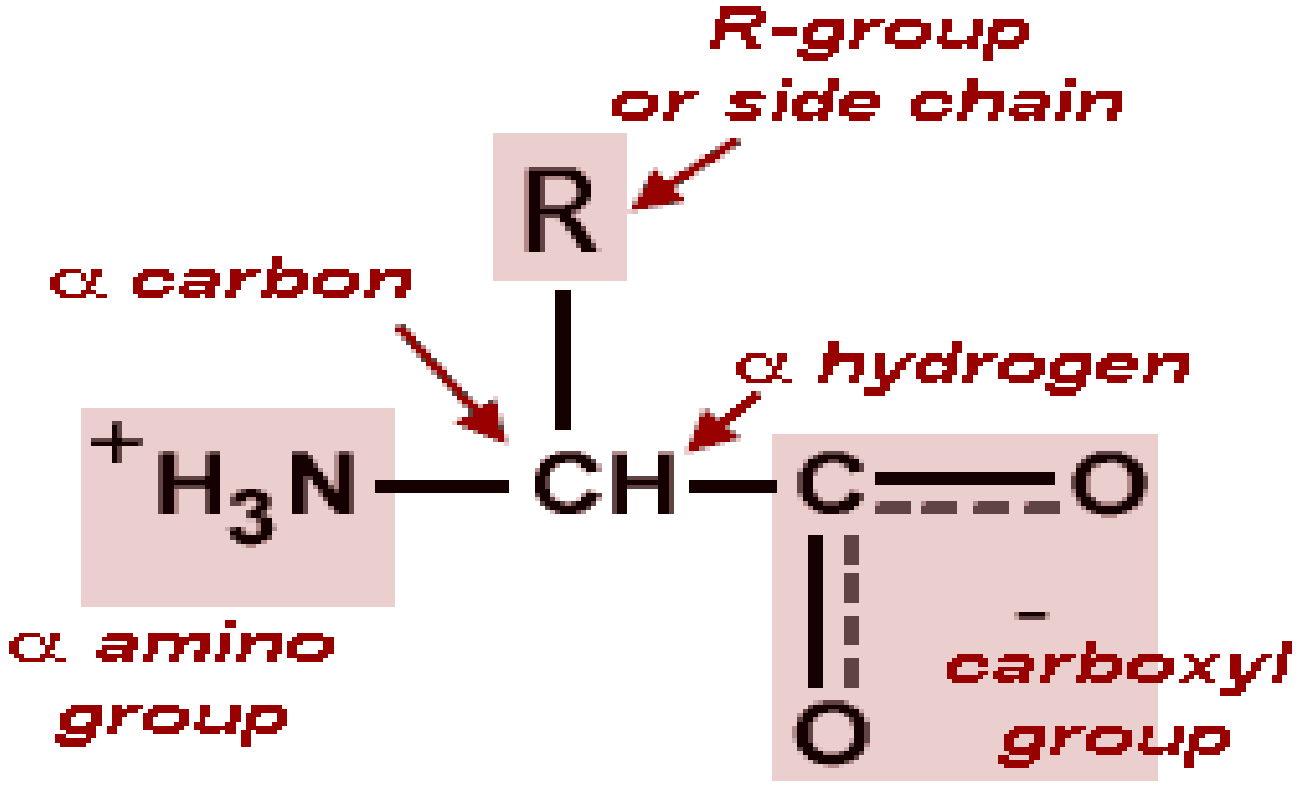
Formation metabolism and elimination of Ammonia

Hyperammonemia

REMOVAL OF NITROGEN FROM AMINO ACIDS

- The presence of the α-amino group keeps amino acids safely locked away from oxidative breakdown.
- Removing the α-amino group is essential for producing energy from any amino acid, and is an obligatory step in the catabolism of all amino acids.
- Once removed, this nitrogen can be incorporated into other compounds or excreted, with the carbon skeletons being metabolized.
- This section describes transamination and oxidative deamination—reactions that ultimately provide ammonia and aspartate, the two sources of urea nitrogen


R-group
or side chain
R
α carbon
α hydrogen
+H3N
CH
C
O
O
–
α amino
group
carboxyl
group

A. Transamination: the funneling of amino groups to glutamate

- The first step in the catabolism of most amino acids is the transfer of their α-amino group to α-ketoglutarate.
- The products are an α-keto acid (derived from the original amino acid) and glutamate.
- α-Ketoglutarate plays a pivotal role in amino acid metabolism by accepting the amino groups from most amino acids, thus becoming glutamate

- Glutamate produced by trans amination can be oxidatively deaminated (see below), or used as an amino group donor in the synthesis of nonessential amino acids.
- This transfer of amino groups from one carbon skeleton to another is catalyzed by a family of enzymes called aminotransferases (formerly called transaminases).
- These enzymes are found in the cytosol and mitochondria of cells through out the body—especially those of the liver, kidney, intestine, and muscle.

B. Glutamate dehydrogenase: the oxidative deamination of amino acids

- In contrast to transamination reactions that transfer amino groups, oxidative deamination by glutamate dehydrogenase results in the liberation of the amino group as free ammonia (NH3)
- These reactions occur primarily in the liver and kidney. They provide α-keto acids that can enter the central pathway of energy metabolism, and ammonia, which is a source of nitrogen in urea synthesis.

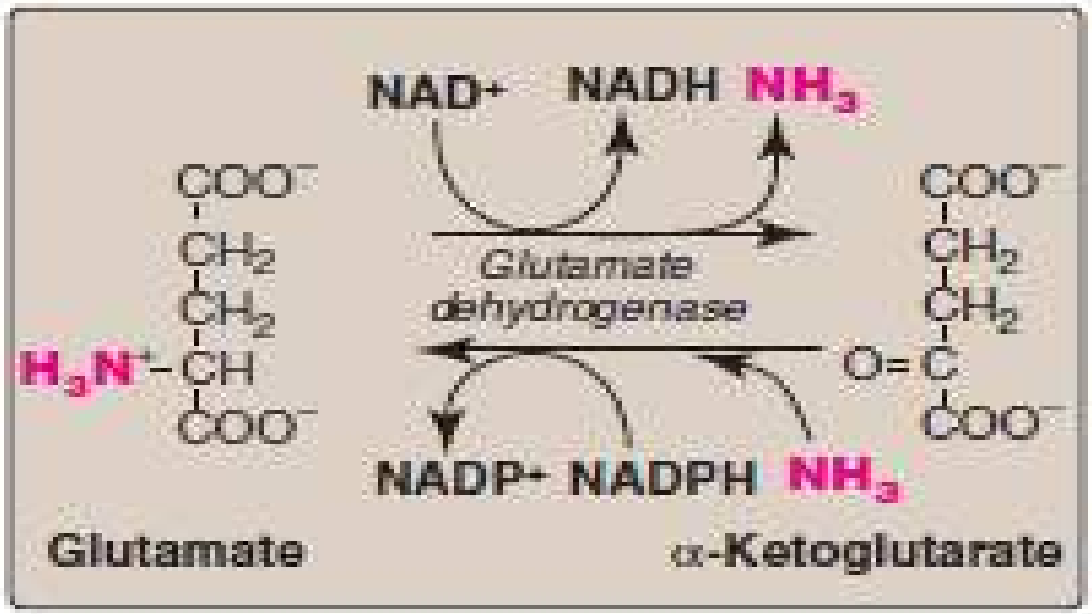


Figure 19.11
Oxidative deamination by *glutamate dehydrogenase*.

C. Transport of ammonia to the liver

- Two mechanisms are available in humans for the transport of ammonia from the peripheral tissues to the liver for its ultimate conversion to urea.
- The first, found in most tissues, uses glutamine synthetase to combine ammonia (NH_3) with glutamate to form glutamine—a nontoxic transport form of ammonia

- The glutamine is transported in the blood to the liver where it is cleaved by glutaminase to produce glutamate and free ammonia.
- The second transport mechanism, used primarily by muscle, involves transamination of pyruvate (the end product of aerobic glycolysis) to form alanine.
- Alanine is transported by the blood to the liver, where it is converted to pyruvate, again by transamination.
- In the liver, the pathway of gluconeogenesis can use the pyruvate to synthesize glucose, which can enter the blood and be used by muscle—a pathway called the glucose-alanine cycle.

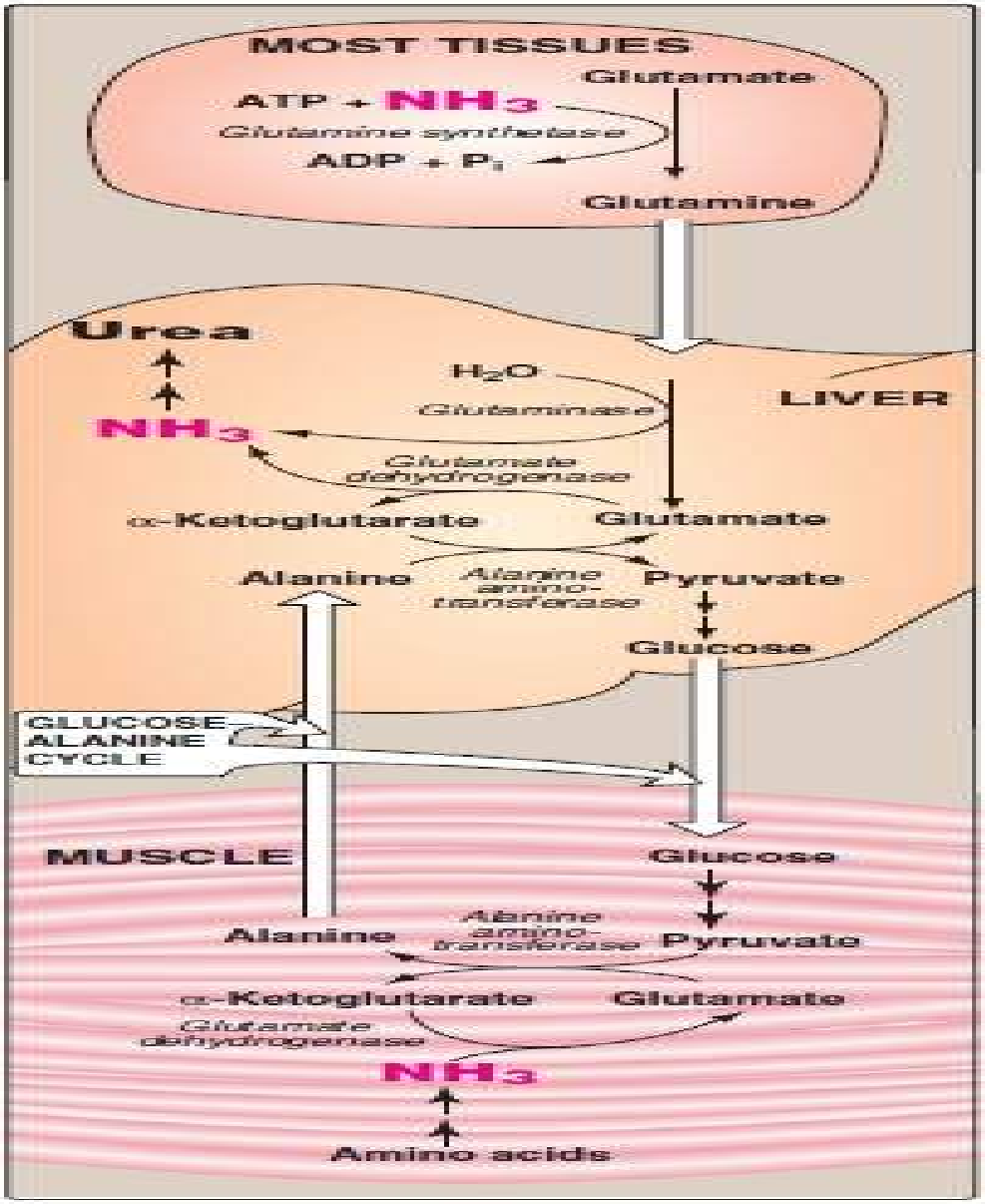


Figure 19.13
Transport of ammonia from peripheral tissues to the liver.

- Urea biosynthesis occurs in four stages:

(1) transamination,

(2) oxidative deamination of glutamate,

(3) ammonia transport, and

(4) reactions of the urea cycle

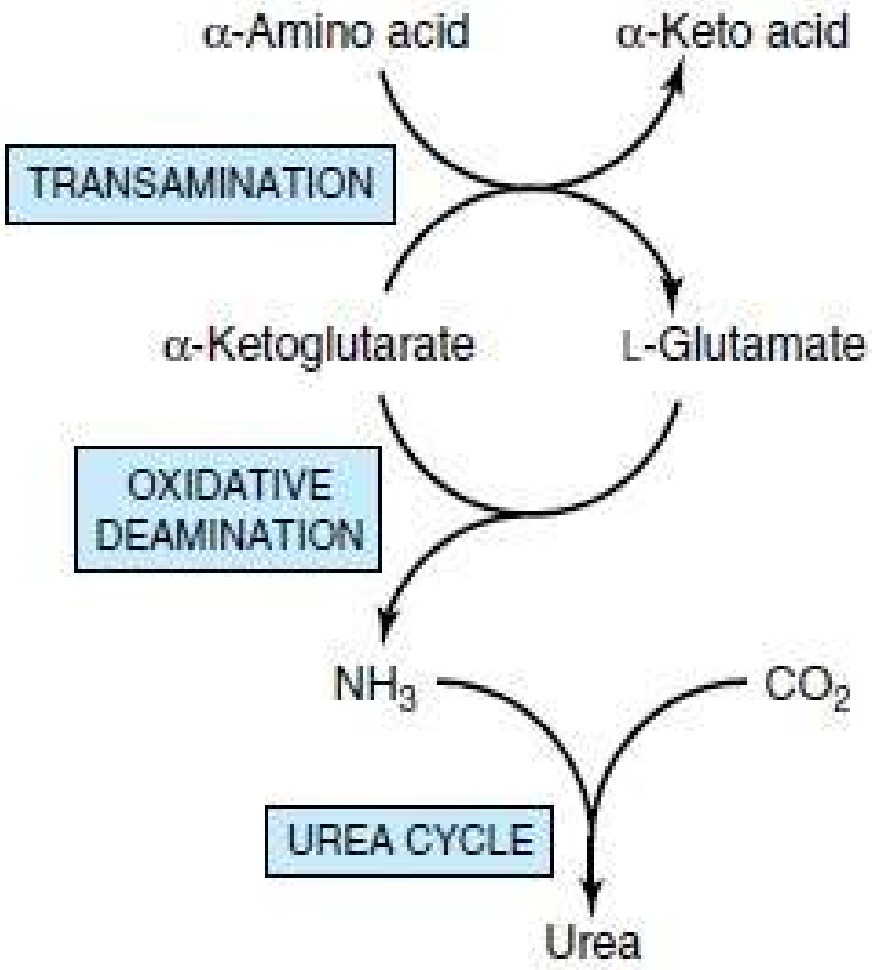


Figure 29–2. Overall flow of nitrogen in amino acid catabolism.

A. Reactions of the cycle

- The first two reactions leading to the synthesis of urea occur in the mitochondria, whereas the remaining cycle enzymes are located in the cytosol (Figure.
- [Note: Gluconeogenesis and heme synthesis also involve both the mitochondrial matrix and the cytosol.]

1. Formation of carbamoyl phosphate

- Formation of carbamoyl phosphate by carbamoyl phosphate synthetase I is driven by cleavage of two molecules of ATP. Ammonia incorporated into carbamoyl phosphate is provided primarily by the oxidative deamination of glutamate by mitochondrial glutamate dehydrogenase. Ultimately, the nitrogen atom derived from this ammonia becomes one of the nitrogens of urea. Carbamoyl phosphate synthetase I requires N-acetylglutamate as a positive allosteric activator (see Figure 19.14). [Note: Carbamoyl phosphate synthetase II participates in the biosynthesis of pyrimidines (see p. 302). It does not require N-acetylglutamate, uses glutamine as the nitrogen source, and occurs in the cytosol.]

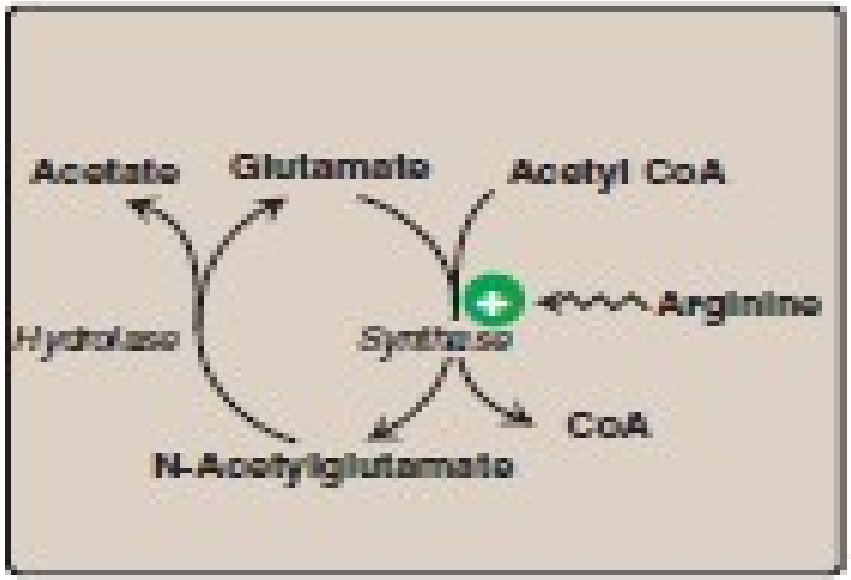


Figure 19.16
Formation and degradation of N-acetylglutamate, an allosteric activator of *carbamoyl phosphate synthetase I*.

2. Formation of citrulline

- The carbamoyl portion of carbamoyl phosphate is transferred to ornithine by ornithine transcarbamoylase (OTC) as the high-energy phosphate is released as Pi.
- The reaction product, citrulline, is transported to the cytosol.
- [Note: Ornithine and citrulline are basic amino acids that participate in the urea cycle, moving across the inner mitochondrial membrane via a cotransporter.
- They are not incorporated into cellular proteins because there are no codons for these amino acids.]
- Ornithine is regenerated with each turn of the urea cycle, much in the same way that oxaloacetate is regenerated by the reactions of the citric acid cycle

Synthesis of argininosuccinate:

- Argininosuccinate synthetase combines citrulline with aspartate to form argininosuccinate.
- The α-amino group of aspartate provides the second nitrogen that is ultimately incorporated into urea. The formation of argininosuccinate is driven by the cleavage of ATP to adenosine monophosphate (AMP) and pyrophosphate.
- This is the third and final molecule of ATP consumed in the formation of urea.

Cleavage of argininosuccinate

- Argininosuccinate is cleaved by argininosuccinate lyase to yield arginine and fumarate. The arginineformed by this reaction serves as the immediate precursor of urea.
- Fumarate produced in the urea cycle is hydrated to malate,providing a link with several metabolic pathways. For example, the malate can be transported into the mitochondria via the malate shuttle, reenter the tricarboxylic acid cycle, and get oxidized to oxaloacetate (OAA), which can be used for gluconeogenesis
- Alternatively, the OAA can be converted to aspartate via transamination , and can enter the urea cycle

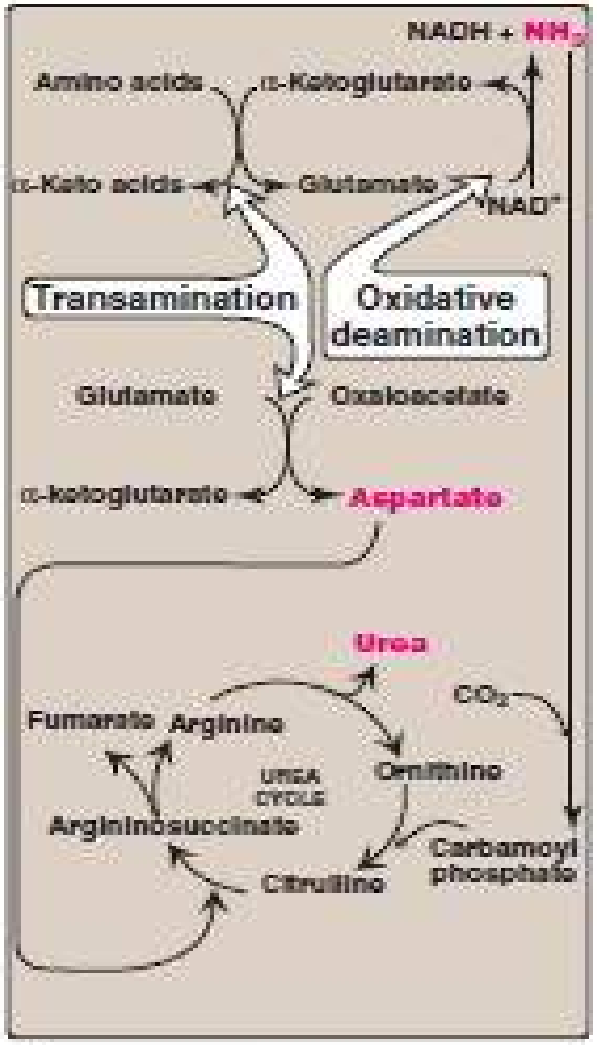
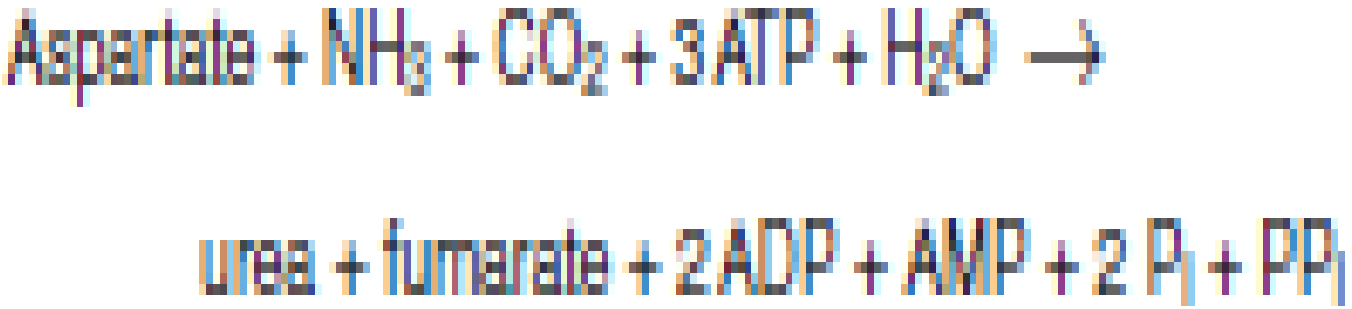


Figure 19.15
Flow of nitrogen from amino acids to urea. Amino groups for urea synthesis are collected in the form of ammonia and aspartate.

Cleavage of arginine to ornithine and urea

- Arginase cleaves arginine to ornithine and urea, and occurs almost exclusively in the liver.
- Thus, whereas other tissues, such as the kidney, can synthesize arginine by these reactions, only the liver can cleave arginine and, thereby, synthesize urea.

Overall stoichiometry of the urea cycle



Stoichometry contd

- Four high-energy phosphate bonds are consumed in the synthesis of each molecule of urea; therefore, the synthesis of urea is irreversible, with a large, negative ΔG. [Note:
- The ATP is replenished by oxidative phosphorylation.] One nitrogen of the urea molecule is supplied by free NH3, and the other nitrogen by aspartate.
- Glutamate is the immediate precursor of both ammonia (through oxidative deamination by glutamate dehydrogenase) and aspartate nitrogen (through transamination of oxaloacetate by AST).
- In effect, both nitrogen atoms of urea arise from glutamate, which, in turn, gathers nitrogen from other amino acids

Regulation of the urea cycle

- N-Acetylglutamate is an essential activator for carbamoyl phosphate synthetase I—the rate-limiting step in the urea cycle.
- N-Acetylglutamate is synthesized from acetyl coenzyme A and glutamate by N-acetylglutamate synthase , in a reaction for which arginine is an activator.
- Therefore, the intrahepatic concentration of N -acetylglutamate increases after ingestion of a protein-rich meal, which provides both a substrate (glutamate) and the regulator of N-acetylglutamate synthesis. This leads to an increased rate of urea synthesis.

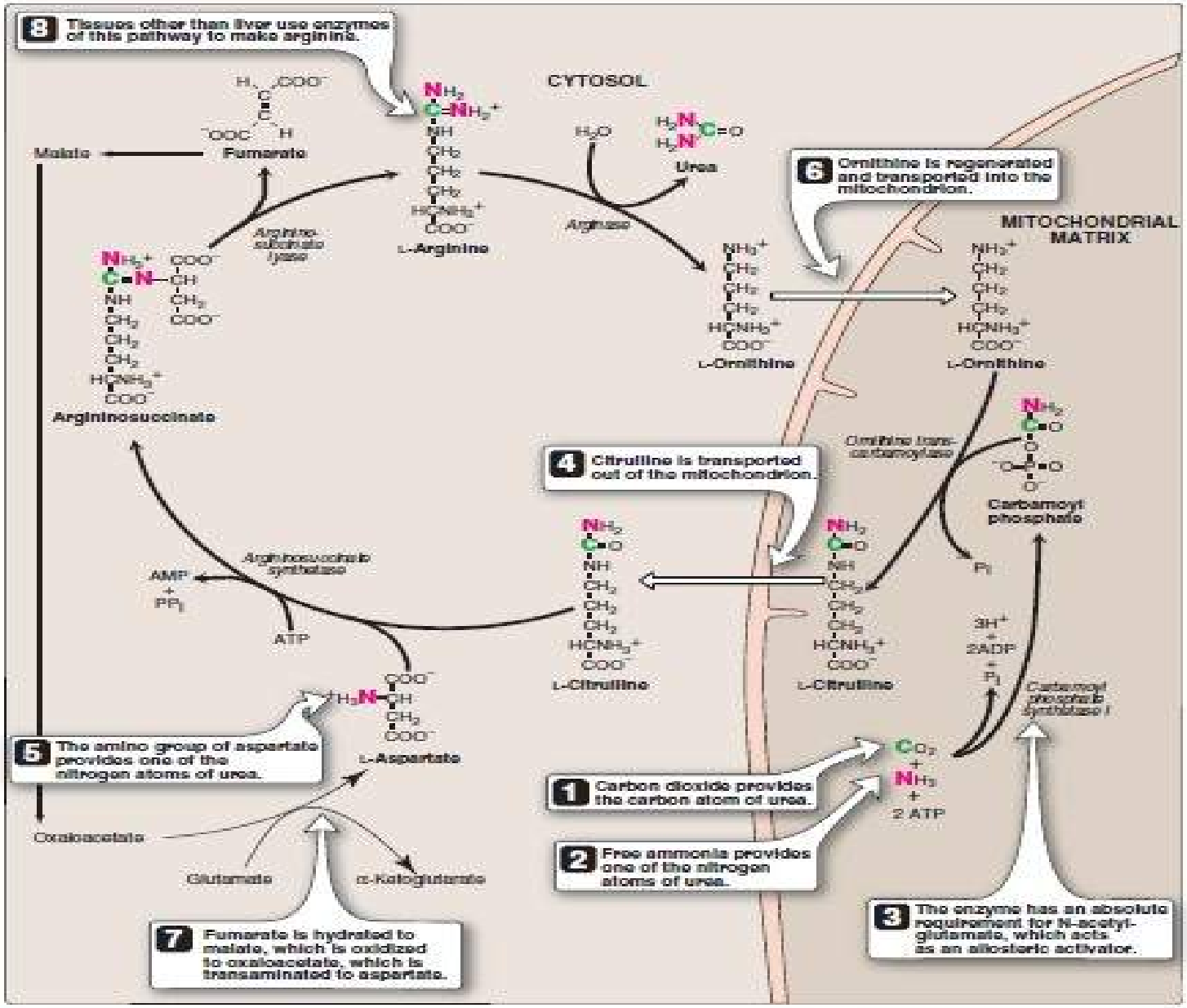


Figure 19.14
Reactions of the urea cycle.

Fate of urea

- Urea diffuses from the liver, and is transported in the blood to the kidneys, where it is filtered and excreted in the urine
- A portion of the urea diffuses from the blood into the intestine, and is cleaved to CO_2 and NH_3 by bacterial urease.
- This ammonia is partly lost in the feces, and is partly reabsorbed into the blood.

- In patients with kidney failure, plasma urea levels are elevated, promoting a greater transfer of urea from blood into the gut.
- The intestinal action of urease on this urea becomes a clinically important source of ammonia, contributing to the hyperammonemia often seen in these patients.
- Oral administration of neomycin reduces the number of intestinal bacteria responsible for this NH3 production.

METABOLISM OF AMMONIA

- Ammonia is produced by all tissues during the metabolism of a variety of compounds, and it is disposed off primarily by formation of urea in the liver.
- However, the level of ammonia in the blood must be kept very low,because even slightly elevated concentrations (hyperammonemia) are toxic to the central nervous system (CNS).

- There must, therefore, be a metabolic mechanism by which nitrogen is moved from peripheral tissues to the liver for ultimate disposal as urea, while at the same time maintaining low levels of circulating ammonia.

A. Sources of ammonia

- **From glutamine**: The kidneys generate ammonia from glutamine by the actions of renal glutaminase (Figure 19.17) and glutamate dehydrogenase.
- Most of this ammonia is excreted into the urine as NH_4^+, which provides an important mechanism for maintaining the body's acid-base balance through the excretion of protons.
- Ammonia is also obtained from the hydrolysis of glutamine by intestinal glutaminase. The intestinal mucosal cells obtain glutamine either from the blood or from digestion of dietary protein.
- [Note: Intestinal glutamine metabolism produces citrulline,which travels to the kidney and is used to synthesize arginine.]

- **From bacterial action in the intestine**: Ammonia is formed from urea by the action of bacterial urease in the lumen of the intestine. This ammonia is absorbed from the intestine by way of the portal vein and is almost quantitatively removed by the liver via conversion to urea.
- **From amines**: Amines obtained from the diet, and monoamines that serve as hormones or neurotransmitters, give rise to ammonia by the action of amine oxidase.
- **From purines and pyrimidines**: In the catabolism of purines and pyrimidines, amino groups attached to the rings are released as ammonia

B. Transport of ammonia in the circulation

- Although ammonia is constantly produced in the tissues, it is pres ent at very low levels in blood. This is due both to the rapid removal of blood ammonia by the liver, and the fact that many tissues, particularly muscle, release amino acid nitrogen in the form of glutamine or alanine, rather than as free ammonia

- **Urea**: Formation of urea in the liver is quantitatively the most important disposal route for ammonia. Urea travels in the blood from the liver to the kidneys, where it passes into the glomerular filtrate.
- **Glutamine**: This amide of glutamic acid provides a nontoxic storage and transport form of ammonia (Figure 19.18). The ATP requiring formation of glutamine from glutamate and ammonia by glutamine synthetase occurs primarily in the muscle and liver, but is also important in the CNS where it is the major mechanism for the removal of ammonia in the brain.
- Glutamine is found in plasma at concentrations higher than other amino acids—a finding consistent with its transport function. Circulating glutamine is removed by the liver and the kidneys and deaminated by glutaminase. In the liver, the NH3 produced is detoxified through conversion to urea, and in the kidney it can be used in the excretion of protons.

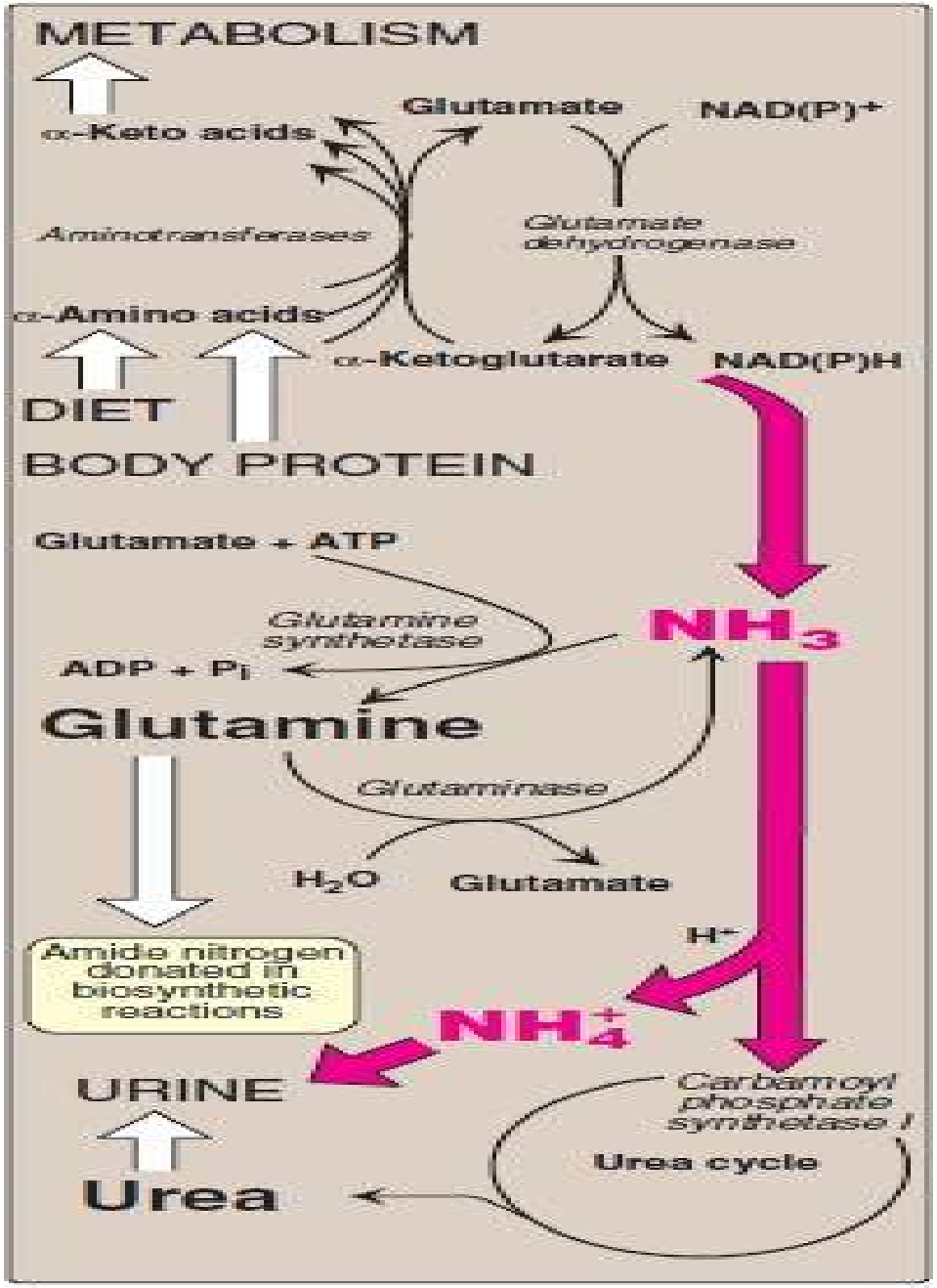


Figure 19.19

Metabolism of ammonia. Urea content in the urine is reported as urinary urea nitrogen or UUN. Urea in blood is reported as BUN (blood urea nitrogen).The enzymes *glutamate dehydrogenase*, *glutamine synthetase*, and *carbamoyl phosphate synthetase I* fix ammonia (NH_3) into organic molecules.

C. Hyperammonemia

- The capacity of the hepatic urea cycle exceeds the normal rates of ammonia generation, and the levels of serum ammonia are normally low (5–35 µmol/L).
- However, when liver function is compromised, due either to genetic defects of the urea cycle or liver disease, blood levels can rise above 1,000 µmol/L.
- Such hy per ammon emia is a medical emergency, because ammonia has a direct neurotoxic effect on the CNS. For example, elevated concentrations of ammonia in the blood cause the symptoms of ammonia intoxication, which include tremors, slurring of speech, somnolence, vomiting, cerebral edema,and blurring of vision.
    - At high concentrations, ammonia can cause coma and death.

The two major types of hyperammonemia are:

- **Acquired hyperammonemia**: Liver disease is a common cause of hyperammonemia in adults, and may be due, for example, to viral hepatitis or to hepatotoxins such as alcohol.
- Cirrhosis of the liver may result in formation of collateral circulation around the liver. As a result, portal blood is shunted directly into the systemic circulation and does not have access to the liver.
- The conversion of ammonia to urea is, therefore, severely impaired,leading to elevated levels of ammonia.

Congenital hyperammonemia:

- Genetic deficiencies of each of the five enzymes of the urea cycle have been described, with an overall prevalence estimated to be 1:25,000 live births.
- Ornithine transcarbamoylase deficiency, which is X-linked, is the most common of these disorders, predominantly affecting males, although female carriers may become symptomatic.

- All of the other urea cycle disorders follow an autosomal recessive inheritance pattern.
- In each case, the failure to synthesize urea leads to hyperammonemia during the first weeks following birth. Historically, urea cycle defects had high morbidity (neurological manifestations) and mortality.

- Treatment included restriction of dietary protein in the presence of sufficient calories to prevent catabolism.
- Administration of compounds that bind covalently to amino acids, producing nitrogen-containing molecules that are excreted in the urine, has improved survival.
- For example, phenylbutyrate given orally is converted to phenylacetate.
- This condenses with glutamine to form phenyl acetylglutamine, which is excreted

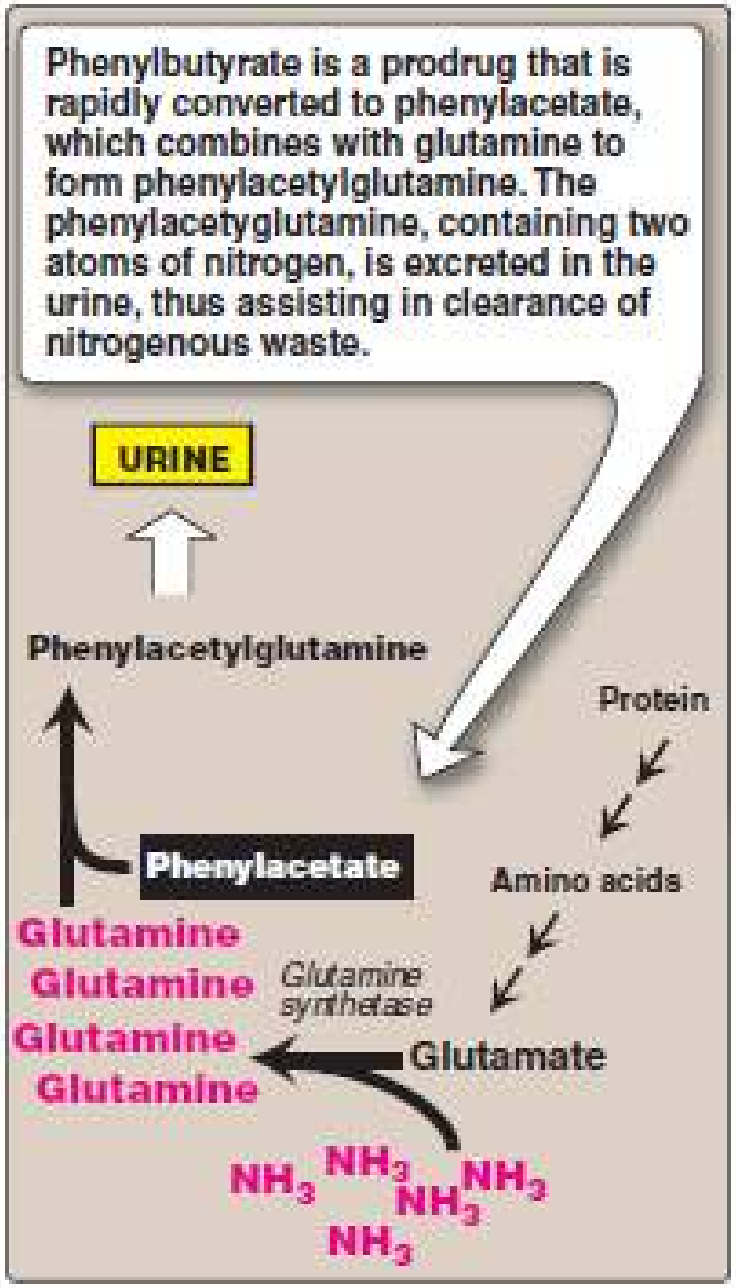


Figure 19.20

Treatment of patients with urea cycle defects by administration of phenylbutyrate to aid in excretion of ammonia.

- **LIPPINCOTTS**